HIER: Heterogeneous Information Bottleneck and Expert Routing for Social Bot Detection

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***Abstract*—Social bots constitute a substantial fraction of active accounts on digital platforms, fundamentally threatening information authenticity and democratic discourse. Contemporary detection methods confront critical limitations: information imbalance across heterogeneous relations, computational challenges in processing massive neighborhoods, and inadequate multi-scale representation learning. We propose HIER (Heterogeneous Information Bottleneck and Expert Routing), a pioneering framework that integrates variational information theory with mixture-of-experts paradigms for social network analysis. HIER introduces relation-aware variational information bottleneck for optimal compression across relationship types, dynamic sparse expert routing that extends mixture-of-experts to edge-level graph processing, and dual-scale mutual information maximization enhancing representation discriminability through neighborhood consistency and graph-level contrastive learning. Experimental validation demonstrates HIER's superior performance across real-world datasets, establishing new benchmarks for heterogeneous social bot detection.**

***Index Terms*—Expert routing, information bottleneck, mutual information, social bot detection.**

I. Introduction

WITH the rapid expansion of online platforms, social bots have become pervasive across digital ecosystems. These sophisticated automated accounts employ advanced behavioral mimicry and are extensively exploited to spread disinformation [1], [2], manipulate public discourse [3], [4], influence political elections [5], [6], [7], and even destabilize societal trust [8], [9]. The growing scale and impact of such accounts pose serious challenges to the credibility and stability of online environments. Consequently, developing effective and robust bot detection methods has become an urgent and essential task in the field of social network analysis.

Early bot detection methods rely on manually engineered features derived from user metadata [10], [11], [12] and processed by traditional classifiers. However, such heuristic strategies often suffer from limited generalization capability. With the advancement of natural language processing techniques, content-based approaches [13], [14], [15] exploit linguistic signals to aid detection, such as semantic features and pragmatic patterns from textual data. Nevertheless, social bots continuously evolve their linguistic behaviors to evade such methods. To capture richer contextual signals, graph neural networks (GNNs) are widely used to integrate users' social connections with textual representations through neighborhood aggregation [16], [17], [18], [19], [20]. Information bottleneck (IB) principles have been extensively studied in deep learning [21], [22], [23], demonstrating compression-discrimination trade-offs in neural networks and hierarchical multimodal learning. We adapt these principles to heterogeneous graphs with relation-specific compression. Complementary detection work addresses rumor propagation [24], [25] and misinformation verification [26], [27], highlighting the broader challenge of combating automated deception in social networks.

Although prior methods have made strides, social bot detection still faces critical challenges: (1) heterogeneous relation types (e.g., following, interaction, retweeting) exhibit varying information quality, obscuring meaningful signals with relational noise; (2) existing aggregation methods fail to distinguish essential information from redundant messages in large-scale neighborhoods, causing over-smoothing; (3) capturing multi-granularity behavioral patterns, from individual anomalies to coordinated network-level campaigns, remains difficult.

To address these challenges, we propose HIER (Heterogeneous Information Bottleneck and Expert Routing), a unified framework integrating three synergistic innovations: relation-aware variational information bottleneck for optimal compression across heterogeneous relations, dynamic sparse expert routing for adaptive edge-level neighbor selection, and dual-scale mutual information maximization for enhanced discriminability. Our main contributions are:

- We propose HIER, a novel framework introducing relation-aware variational information bottleneck that automatically learns optimal compression policies across heterogeneous relationship types in social networks.
- We develop dynamic sparse expert routing extending mixture-of-experts to edge-level graph processing, and dual-scale mutual information maximization enhancing local-global representation discriminability.
- Comprehensive experiments demonstrate HIER's superior performance over state-of-the-art methods, with significant improvements in heterogeneous scenarios and effective handling of class imbalance.

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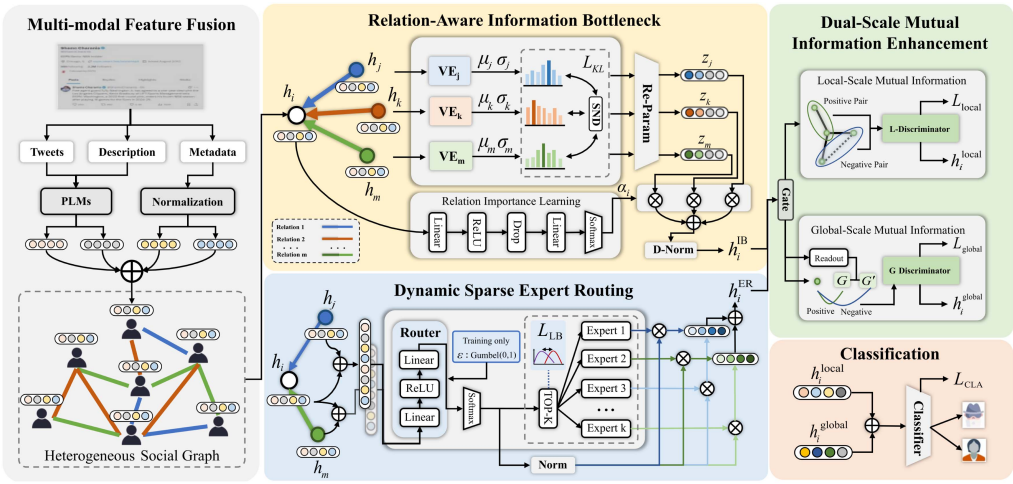

Fig. 1. The architecture of HIER. The framework processes multi-modal features through three components: relation-aware variational information bottleneck, dynamic sparse expert routing, and dual-scale mutual information maximization.

II. Methodology

In this section, we will introduce HIER, a novel framework for social bot detection. As shown in Fig. 1, HIER integrates three core collaborative components to address the challenges in heterogeneous social networks.

A. Multi-Modal Feature Fusion

Social media users exhibit diverse characteristics across multiple modalities. We extract user representations from four primary feature modalities to construct comprehensive node embeddings.

For each user node v_i, tweet content and profile descriptions are processed through pre-trained language models (PLMs) to obtain semantic embeddings $\mathbf{x}_i^{tweet} \in \mathbb{R}^{d_t}$ and $\mathbf{x}_i^{desc} \in \mathbb{R}^{d_d}$ respectively. Account metadata is divided into categorical attributes (e.g., verification status) normalized as $\mathbf{x}_i^{cat} \in \mathbb{R}^{d_c}$ and numerical statistics (e.g., follower counts) standardized as $\mathbf{x}_i^{num} \in \mathbb{R}^{d_n}$. The initial node representation is obtained through direct concatenation of all feature modalities:

$$\mathbf{h}_i = [\mathbf{x}_i^{tweet} \oplus \mathbf{x}_i^{desc} \oplus \mathbf{x}_i^{cat} \oplus \mathbf{x}_i^{num}] \in \mathbb{R}^d. \tag{1}$$

B. Relation-Aware Information Bottleneck

Social networks contain relations of varying information quality and noise levels. To address this heterogeneity, we extend variational information bottleneck principles [28] to learn optimal compression strategies for different relations.

For each relation type r_k, we employ a dedicated variational encoder VE_k that learns to compress neighbor representations while preserving task-relevant information:

$$[\boldsymbol{\mu}_k(\mathbf{h}_j); \log \boldsymbol{\sigma}_k^2(\mathbf{h}_j)] = \mathrm{VE}_k(\mathbf{h}_j), \tag{2}$$

where $\mathbf{h}_j$ represents the source node representation, and $\boldsymbol{\mu}_k(\mathbf{h}_j) \in \mathbb{R}^{d'}, \boldsymbol{\sigma}_k^2(\mathbf{h}_j) \in \mathbb{R}^{d'}$ are the learned mean and variance parameters for relation r_k.

For each relation r_k, we perform variational sampling using the reparameterization trick and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathbf{z}_{j\to i}^{(k)} = \boldsymbol{\mu}_k(\mathbf{h}_j) + \boldsymbol{\epsilon} \odot \exp(\tau \cdot \log \boldsymbol{\sigma}_k^2(\mathbf{h}_j)), \tag{3}$$

where $\odot$ denotes element-wise multiplication.

To capture node-specific relation importance, we learn adaptive weights $\boldsymbol{\alpha}_i = \mathrm{softmax}(\mathrm{MLP}(\mathbf{h}_i))$ through the Relation Importance Learning module. The final representation with degree normalization (D-Norm) is:

$$\mathbf{h}_i^{IB} = \frac{1}{\max(|\mathcal{N}_i|, 1)} \sum_{k=1}^{K} \sum_{j \in \mathcal{N}_i^{(k)}} \alpha_{i,k} \cdot \mathbf{z}_{j\to i}^{(k)}, \tag{4}$$

where $\mathcal{N}_i^{(k)}$ represents the neighbor set of target node i under relation r_k.

Information compression is enforced through KL divergence with a standard Gaussian prior:

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|\mathcal{E}_k|} \sum_{(i,j)\in\mathcal{E}_k} \mathrm{KL}(q(\mathbf{z}^{(k)}|\mathbf{h}_j) \| \mathcal{N}(\mathbf{0}, \mathbf{I})). \tag{5}$$

Theoretical Justification: Standard information bottleneck applies uniform compression, which is suboptimal when relations have heterogeneous information quality. For each relation r_k, we optimize:

$$\min_{q_k} \mathcal{L}_{\mathrm{IB}}^{(k)} = I(Z^{(k)}; H_j) - \beta I(Z^{(k)}; Y), \tag{6}$$

where $I(Z^{(k)}; Y)$ maintains task-relevant information with labels Y, while the KL term in (5) bounds compression $I(Z^{(k)}; H_j)$. This enables relation-specific compression rates: high-quality relations preserve more information ($\sigma_k \to 0$), while noisy relations apply aggressive filtering (large σ_k). Relation-specific compression can theoretically reduce generalization error compared to uniform compression in heterogeneous settings where relation quality variance $\mathrm{Var}(\rho_k) > 0$.

C. Dynamic Sparse Expert Routing

To handle diverse node interaction patterns efficiently, we propose dynamic sparse expert routing that adaptively selects specialized experts for each edge-level interaction.

Given target node $\mathbf{h}_i$ and source nodes $\{\mathbf{h}_j, \mathbf{h}_m, \ldots\}$, the router network computes expert selection scores for each edge

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interaction. For edge $(j \to i)$, we construct routing context:

$$\mathbf{c}_{j\to i} = [\mathbf{h}_j \oplus \mathbf{h}_i] \in \mathbb{R}^{2d}. \tag{7}$$

The routing network computes expert preference scores for each edge through a gating mechanism:

$$\mathbf{r}_{j\to i} = \mathrm{MLP}_{\text{router}}\left(\mathbf{c}_{j\to i}\right) \in \mathbb{R}^{M}, \tag{8}$$

where M is the number of expert networks.

To enable differentiable discrete selection during training, we apply Gumbel noise to the routing logits:

$$\tilde{\mathbf{r}}_{j\to i} = \mathbf{r}_{j\to i} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathrm{Gumbel}(0,1). \tag{9}$$

We then apply temperature-scaled softmax followed by top-k sparse selection:

$$(\boldsymbol{\alpha}_{j\to i}, \mathbf{T}_{j\to i}) = \text{ Top- } k\left(\mathrm{softmax}\left(\frac{\tilde{\mathbf{r}}_{j\to i}}{\tau}\right), k\right), \tag{10}$$

where $\boldsymbol{\alpha}_{j\to i}$ contains normalized weights for selected experts, and $\mathbf{T}_{j\to i} \in \mathbb{R}^k$ contains the corresponding expert indices.

The routed message for edge $(j \to i)$ is computed as a weighted combination of the top-k selected experts. Each expert is implemented as a two-layer feed-forward network:

$$\mathrm{Expert}_m\left(\mathbf{c}_{j\to i}\right) = \mathbf{W}_m^{(2)} \cdot \mathrm{ReLU}\left(\mathbf{W}_m^{(1)} \cdot \mathbf{c}_{j\to i} + \mathbf{b}_m^{(1)}\right) + \mathbf{b}_m^{(2)}, \tag{11}$$

where $\mathbf{W}_m^{(1)} \in \mathbb{R}^{d\times 2\tilde{}d}, \mathbf{W}_m^{(2)} \in \mathbb{R}^{d\times d}$ are expert-specific parameters. We employ M experts, each with independent weights enabling specialization in distinct edge interaction patterns. The final expert routing representation is:

$$\mathbf{h}_i^{\mathrm{ER}} = \frac{1}{|\mathcal{N}_i|} \sum_{j\in\mathcal{N}_i} \sum_{s=1}^{k} \alpha_{j\to i}^{(s)} \cdot \mathrm{Expert}_{T_{j\to i}^{(s)}}(\mathbf{c}_{j\to i}). \tag{12}$$

Load balancing loss $\mathcal{L}_{LB}$ prevents expert collapse by encouraging uniform expert utilization:

$$\mathcal{L}_{\mathrm{LB}} = \frac{1}{M} \sum_{m=1}^{M} \left(\frac{1}{|\mathcal{E}|} \sum_{(i,j)\in\mathcal{E}} \mathbb{I}(m \in \mathbf{T}_{j\to i}) - \frac{1}{M}\right)^2. \tag{13}$$

D. Dual-Scale Mutual Information Maximization

To enhance node representation discriminability, we employ dual-scale mutual information maximization that optimizes both local neighborhood consistency and global graph-level alignment through contrastive learning principles.

We first combine representations from previous modules through a gating mechanism:

$$\mathbf{h}_i^{\mathrm{MI}} = \mathrm{Gate}(\mathbf{h}_i^{\mathrm{IB}}, \mathbf{h}_i^{\mathrm{ER}}). \tag{14}$$

At the local scale, we construct positive pairs $(\mathbf{h}_i^{\mathrm{MI}}, \mathbf{h}_j^{\mathrm{MI}})$ from connected nodes (i,j) and negative pairs $(\mathbf{h}_i^{\mathrm{MI}}, \mathbf{h}_{j'}^{\mathrm{MI}})$ from unconnected nodes (i,j'). The local mutual information enhances representations $\mathbf{h}_i^{\mathrm{local}}$ through discriminator D_{local}:

$$\begin{aligned}\mathcal{L}_{\mathrm{local}} = &-\mathbb{E}_{(i,j)\in\mathcal{E}}[\log\sigma(D_{\mathrm{local}}([\mathbf{h}_i^{\mathrm{MI}}; \mathbf{h}_j^{\mathrm{MI}}]))] \\ &- \mathbb{E}_{(i,j')\notin\mathcal{E}}[\log(1-\sigma(D_{\mathrm{local}}([\mathbf{h}_i^{MI}; \mathbf{h}_{j'}^{\mathrm{MI}}])))]. \end{aligned} \tag{15}$$

At the global scale, we compute graph representation $\mathbf{g} = \mathrm{READOUT}(\{\mathbf{h}_1^{\mathrm{MI}}, \ldots, \mathbf{h}_N^{\mathrm{MI}}\})$ and construct positive pairs $(\mathbf{h}_i^{\mathrm{MI}}, \mathbf{g})$ with the original graph and negative pairs $(\mathbf{h}_i^{\mathrm{MI}}, \mathbf{g}')$ with corrupted representations. The global contrastive learning produces $\mathbf{h}_i^{\mathrm{global}}$ and the combined mutual information loss is:

$$\mathcal{L}_{\mathrm{MI}} = \mathcal{L}_{\mathrm{local}} + \lambda_{\mathrm{global}}\mathcal{L}_{\mathrm{global}}. \tag{16}$$

E. Model Integration and Classification

The dual-scale enhanced representations are directly concatenated to form the final input for classification, where $\mathbf{h}_i^{\mathrm{final}} = [\mathbf{h}_i^{\mathrm{local}} \oplus \mathbf{h}_i^{\mathrm{global}}]$ combines both neighborhood consistency features and graph-level pattern representations.

The classification is performed through a two-layer neural network that processes the enhanced representations:

$$\hat{\mathbf{y}}_i = \mathrm{softmax}(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1 \cdot \mathbf{h}_i^{\mathrm{final}} + \mathbf{b}_1) + \mathbf{b}_2). \tag{17}$$

The classification loss for labeled nodes is computed as standard cross-entropy:

$$\mathcal{L}_{\mathrm{CLA}} = -\frac{1}{|\mathcal{V}_L|} \sum_{i\in\mathcal{V}_L} \sum_{c=1}^{C} y_{i,c} \log \hat{y}_{i,c}, \tag{18}$$

where $\mathcal{V}_L$ denotes the set of labeled nodes, C is the number of classes. The complete HIER framework is optimized through joint loss that combines classification objective with auxiliary regularization terms:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{CLA}} + \lambda_{\mathrm{KL}}\mathcal{L}_{\mathrm{KL}} + \lambda_{\mathrm{LB}}\mathcal{L}_{\mathrm{LB}} + \lambda_{\mathrm{MI}}\mathcal{L}_{\mathrm{MI}}. \tag{19}$$

III. Experiments

A. Dataset and Parameter Setting

We evaluate our proposed method on two widely used bot detection benchmarks: Cresci-15 and MGTAB. Cresci-15 consists of 5,301 users (1,950 humans and 3,351 bots), with a total of over 2.8 million tweets and 7 million user-user interaction edges. Human users account for 2.63 million tweets, while bots generate 196 K tweets. MGTAB contains 10,199 users (7,451 humans and 2,748 bots), with approximately 1.7 million edges and 7 different relation types.

We train for 200 epochs using the Adam optimizer [29] with a learning rate of 1e-3. The dropout rate is set to 0.5 to prevent overfitting. All experiments are conducted on an NVIDIA H20 GPU with 96 GB memory.

B. Comparison With State-of-the-Art Models

Table I presents comprehensive evaluation results on Cresci-15 and MGTAB datasets. HIER consistently outperforms all baselines across evaluation metrics with statistically significant improvements. Moreover, the relatively small standard deviations across five runs indicate that HIER is training-stable and robust compared with contrastive baselines such as GRACE.

On Cresci-15, HIER achieves 99.36% accuracy and 99.29% F1-score, outperforming the strongest baseline BotMoE by 0.88% accuracy. HIER's superior performance stems from relation-aware information bottleneck that filters noisy interactions while preserving discriminative signals, and dynamic sparse expert routing that enables adaptive neighbor selection without over-smoothing. Compared to relation-aware baselines (BotRGCN, RGT), HIER's variational compression provides more principled heterogeneous relation handling.

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TABLE I
PERFORMANCE COMPARISON ON CRESCI-15 AND MGTAB IN TERMS OF ACCURACY, F1-SCORE, RECALL AND PRECISION. THE BEST RESULTS ARE HIGHLIGHTED WITH BOLD

Method	Cresci-15				MGTAB			
	Accuracy	F1-score	Recall	Precision	Accuracy	F1-score	Recall	Precision
GCN [30]	94.47±0.73	94.78±0.55	95.41±0.49	93.73±0.69	84.07±1.44	79.31±2.12	79.07±3.48	81.08±1.90
GAT [31]	95.51±0.43	95.72±0.60	95.39±0.83	93.85±0.48	84.12±1.66	76.44±3.79	73.90±4.14	77.59±1.28
GraphSage [32]	95.67±0.39	95.70±0.41	95.48±0.79	95.81±0.32	82.65±5.54	77.70±5.95	82.03±4.71	75.64±3.31
FAGCN [33]	95.88±0.38	95.42±0.29	95.68±0.33	95.37±0.37	88.11±1.43	77.43±3.20	76.36±7.90	79.19±3.41
H2GCN [34]	96.87±0.57	97.04±0.62	97.02±0.79	96.94±0.55	89.09±1.16	79.99±1.53	81.00±5.94	79.70±5.01
BotRGCN [16]	96.52±0.71	97.30±0.53	97.18±0.42	97.43±0.52	87.55±0.61	84.76±0.82	84.22±2.07	84.39±1.03
RGT [17]	97.05±0.59	97.73±0.48	97.57±0.24	97.49±0.37	88.60±1.35	84.96±0.87	84.21±1.25	84.48±1.93
HGT [35]	96.37±0.26	96.78±0.32	96.26±0.44	96.36±0.38	88.54±1.03	85.12±2.09	84.79±3.18	84.93±2.01
DGI [36]	93.70±0.37	92.45±0.61	93.89±0.27	92.88±0.45	87.08±0.98	75.67±1.62	74.61±2.58	76.81±1.56
GRACE [37]	84.18±2.44	83.90±2.79	82.73±3.64	82.59±3.09	83.16±1.60	66.99±3.90	63.83±6.00	70.77±2.02
SEBot [38]	95.83±0.32	95.41±0.29	95.57±0.34	95.39±0.29	90.46±1.44	82.12±2.42	81.73±2.77	82.52±2.19
BotMoE [39]	98.48±0.21	98.82±0.11	98.17±0.19	98.24±0.36	87.45±0.87	78.23±1.31	77.78±2.15	79.18±2.45
OS-GNN [40]	96.65±0.36	96.38±0.39	96.73±0.27	96.55±0.31	86.75±0.74	85.39±0.71	83.34±1.92	83.51±1.38
HIER	**99.36±0.44**	**99.29±0.49**	**99.34±0.37**	**99.23±0.56**	**91.37±0.71**	**88.67±1.01**	**87.92±2.57**	**89.52±1.43**

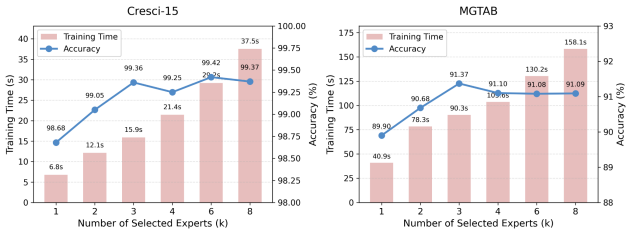

Fig. 2. Impact of the number of selected experts k on classification performance and training efficiency.

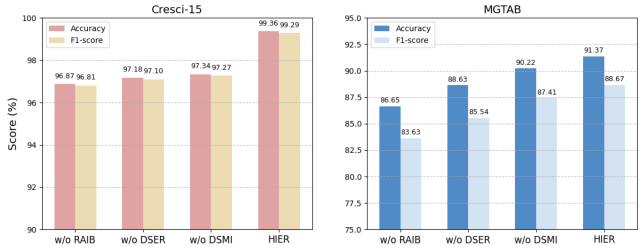

Fig. 3. Ablation study removing each component: RAIB (Relation-Aware Information Bottleneck), DSER (Dynamic Sparse Expert Routing), DSMI (Dual-Scale Mutual Information).

On MGTAB, a challenging dataset with lower homophily and greater class imbalance, HIER achieves 91.37% accuracy and 88.67% F1-score. The substantial improvements over homogeneous methods (GCN: 7.30% accuracy gain) validate the necessity of relation-aware modeling in heterogeneous social networks. HIER surpasses BotMoE by 3.92% F1-score, demonstrating that dynamic sparse routing with adaptive expert selection outperforms static mixture-of-experts. The consistent performance across both datasets demonstrates HIER's robustness to varying graph structures.

HIER's time complexity $O(k \cdot |E| \cdot d + K \cdot |V| \cdot d^2)$ achieves computational efficiency through sparse expert routing and parallelizable relation-specific processing. Compared to dense MoE approaches requiring $O(M \cdot |E| \cdot d)$, HIER reduces expert computations by 62.5%. While exceeding simple baselines like GCN($O(|E| \cdot d)$), the substantial accuracy improvement justifies the modest complexity increase for handling heterogeneous social networks.

These results validate HIER's effectiveness in handling heterogeneous relations and class imbalance in social bot detection. Fig. 2 validates our choice of $k = 3$, balancing accuracy and efficiency across both datasets. Compared to soft attention-based routing which activates all experts, our sparse top-k selection achieves superior performance, demonstrating that hard sparsity enhances expert specialization and prevents redundant activations.

C. Ablation Experiments

To validate each component's contribution, we conduct systematic ablation studies by removing key modules. Fig. 3 demonstrates the necessity of each component. Removing Relation-Aware Information Bottleneck (RAIB) causes the most significant degradation (2.49% on Cresci-15, 4.72% on MGTAB), validating that heterogeneous relations require specialized filtering. Dynamic Sparse Expert Routing (DSER) removal results in moderate drops (1.18%, 2.74%), demonstrating adaptive neighbor selection value. Dual-Scale Mutual Information (DSMI) ablation shows consistent degradation (2.02%, 1.15%).

IV. CONCLUSION

This letter presents HIER, a novel framework that systematically addresses fundamental challenges in heterogeneous social bot detection through relation-aware information bottleneck, dynamic sparse expert routing, and dual-scale mutual information maximization. Comprehensive experiments demonstrate HIER's superior performance across diverse network structures, achieving consistent improvements over state-of-the-art baselines. Beyond performance gains, HIER establishes principled foundations for heterogeneous graph learning by integrating information-theoretic principles with mixture-of-experts paradigms. Future work will focus on scaling to million-node networks, empirical multi-platform validation to assess cross-platform generalization, exploration of parameter-efficient multi-modal fusion strategies, and theoretical analysis of information compression-expert routing interactions.

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